RIG-SMOOTH AND RIG-ETALE DESCENT FOR MORPHISMS OF FORMAL ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 5638-5732, following the statement at 5625-5636.

Editorial note. The 'some details omitted' at the contradiction setup is the instantiation of the cited characterization lemma, flagged in the notes; the rig-etale case is the mirror argument with both cohomology objects vanishing.

Editorial note. Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 5676 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

Allowed standard prerequisites (editorial).

- Characterization of rig-smooth maps by the completed naive cotangent complex. lemma-equivalent-with-artin-smooth (same chapter, cited at ~5671; its own proof has stated omissions and was not counted). Failure of rig-smoothness gives a prime with H^-1 nonzero or H^0 not finite free.
- Flatness of rig-smooth ring maps. lemma-rig-smooth-rig-flat (carded L26159, cited at ~5704). B_q -> A_p is flat, hence faithfully flat.
- Exact sequence of completed naive cotangent complexes. lemma-exact-sequence-NL and lemma-exact-sequence-NL-rig-smooth (same chapter, cited at ~5710). Transmits the vanishing and finite-free conclusions along C -> B -> A.
- DVR lifting through etale coverings. Algebra, Lemma algebra-lemma-exists-dvr (cross-chapter) and lemma-etale-covering-rig-surjective (same chapter, cited at ~5695). Produces the test ring R and lifts the Spf(R)-point to some Spf(A_i).

Source reference. Original reference 00PH: focused-seeds/geometry/S9-restricted-rig-etale-descent/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original definition 0GCP: sources/stacks-restricted/pinned-original-source.tex, source lines 4622–4622. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:restricted.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 0AIM: focused-seeds/geometry/S9-restricted-rig-etale-descent/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0GCR: sources/stacks-restricted/pinned-original-source.tex, source lines 4692–4692. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0AQS: sources/stacks-restricted/pinned-original-source.tex, source lines 5324–5324. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0GAJ: sources/stacks-restricted/pinned-original-source.tex, source lines 843–843. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0AQU: sources/stacks-restricted/pinned-original-source.tex, source lines 5363–5363. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0ALM: sources/stacks-restricted/pinned-original-source.tex, source lines 725–725. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0GH2: sources/stacks-restricted/pinned-original-source.tex, source lines 4561–4561. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0GC0: sources/stacks-restricted/pinned-original-source.tex, source lines 2556–2556. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0GH1: sources/stacks-restricted/pinned-original-source.tex, source lines 4460–4460. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ and $g : Y \to Z$ be morphisms of locally Noetherian formal algebraic spaces over S. Assume*

(1) *g is locally of finite type,*
(2) *f is rig-smooth (resp. rig-étale) and rig-surjective,*
(3) *$g \circ f$ is rig-smooth (resp. rig-étale)*

then g is rig-smooth (resp. rig-étale).

Proof. We will prove this in the rig-smooth case and indicate the necessary changes to prove the rig-étale case at the end of the proof. Consider a commutative diagram

$$\begin{array}{ccccc} X \times_Y V & \longrightarrow & V & \longrightarrow & W \\ \downarrow & & \downarrow & & \downarrow \\ X & \longrightarrow & Y & \longrightarrow & Z \end{array}$$

with V and W affine formal algebraic spaces, $V \to Y$ and $W \to Z$ representable by algebraic spaces and étale. We have to show that $V \to W$ corresponds to a rig-smooth map of adic Noetherian topological rings, see Definition 0GCP. We may write $V = \mathrm{Spf}(B)$ and $W = \mathrm{Spf}(C)$ and that $V \to W$ corresponds to an adic ring map $C \to B$ which is topologically of finite type, see Lemma 0GC0.

We will use below without further mention that $X \times_Y V \to V$ is rig-smooth and rig-surjective, see Lemmas 0GCR and 0AQS. Also, the composition $X \times_Y V \to V \to W$ is rig-smooth since $g \circ f$ is rig-smooth.

Let $I \subset C$ be an ideal of definition. The module Assume $C \to B$ is not rig-smooth to get a contradiction. This means that there exists a prime ideal $\mathfrak{q} \subset B$ not containing IB such that either $H^{-1}(NL^\wedge_{B/C})_\mathfrak{p}$ is nonzero or $H^0(NL^\wedge_{B/C})_\mathfrak{p}$ is not a finite free $B_\mathfrak{q}$-module. See Lemma 0GAJ; some details omitted. We may choose a maximal ideal $IB + \mathfrak{q} \subset \mathfrak{m}$. By Algebra, Lemma 00PH we can find a complete discrete valuation ring R and an injective local ring homomorphism $(B/\mathfrak{q})_\mathfrak{m} \to R$.

After replacing R by an extension, we may assume given a lift $\mathrm{Spf}(R) \to X \times_Y V$ of the adic morphism $\mathrm{Spf}(R) \to V = \mathrm{Spf}(B)$. Choose an étale covering $\{\mathrm{Spf}(A_i) \to X \times_Y V\}$ as in Formal Spaces, Definition 0AIM. By Lemma 0AQU we may assume $\mathrm{Spf}(R) \to X \times_Y V$ lifts to a morphism $\mathrm{Spf}(R) \to \mathrm{Spf}(A_i)$ for some i (this might require replacing R by another extension). Set $A = A_i$. Consider the ring maps

$$C \to B \to A \to R$$

Let $\mathfrak{p} \subset A$ be the kernel of the map $A \to R$ and note that $\mathfrak{p}$ lies over $\mathfrak{q}$. We know that $C \to A$ and $B \to A$ are rig-smooth. In particular the ring map $B_\mathfrak{q} \to A_\mathfrak{p}$ is flat by Lemma 0GH1. Consider the associated exact sequence

$$\begin{array}{l} H^0(NL^\wedge_{B/C}) \otimes_B A_\mathfrak{p} \longrightarrow H^0(NL^\wedge_{A/C})_\mathfrak{p} \longrightarrow H^0(NL^\wedge_{A/B})_\mathfrak{p} \longrightarrow 0 \\ \qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad \nwarrow \\ 0 \longrightarrow H^{-1}(NL^\wedge_{B/C} \otimes_B A)_\mathfrak{p} \longrightarrow H^{-1}(NL^\wedge_{A/C})_\mathfrak{p} \longrightarrow H^{-1}(NL^\wedge_{A/B})_\mathfrak{p} \end{array}$$

of Lemmas 0ALM and 0GH2. Given the rig-smoothness of $C \to A$ and $B \to A$ we conclude that $H^{-1}(NL^\wedge_{B/C} \otimes_B A)_\mathfrak{p} = 0$ and that $H^0(NL^\wedge_{B/C}) \otimes_B A_\mathfrak{p}$ is finite free as a kernel of a surjection of finite free $A_\mathfrak{p}$-modules. Since $B_\mathfrak{q} \to A_\mathfrak{p}$ is flat and hence

faithfully flat, this implies that $H^{-1}(NL^\wedge_{B/C})_\mathfrak{q} = 0$ and that $H^0(NL^\wedge_{B/C})_\mathfrak{q}$ is finite free which is the contradiction we were looking for.

In the rig-étale case one argues in exactly the same manner but the conclusion obtained is that both $H^{-1}(NL^\wedge_{B/C})_\mathfrak{q}$ and $H^0(NL^\wedge_{B/C})_\mathfrak{q}$ are zero. $\square$